

Worksheet on Surds

Laws ($a, b \in \mathbb{R}$ and $n, m \in \mathbb{Z}$)

- $\sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{ab}$
- $\sqrt[n]{\sqrt[n]{a}} \times \sqrt[n]{b} = \sqrt[n]{ab}$
- $(\sqrt[n]{a})^n = \sqrt[n]{a^n}$
- $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$

Exercise 1

Simplify without the use of a calculator

- $2\sqrt{8} - 4\sqrt{32} + 3\sqrt{50}$
- $\sqrt{50}(\sqrt{18} + \sqrt{32})$
- $(2 - \sqrt{3})^2 + (2 + \sqrt{3})^2$
- $\left(\frac{\sqrt{8}}{2} - \sqrt{18} + \sqrt{72}\right)\left(\frac{2}{\sqrt{2}} + \frac{4}{\sqrt{8}}\right)^{-1}$

Exercise 2

Solve x without the use of a calculator

- $162 - 2\sqrt[4]{x^4} = 0$
- $3^{2x} - 3^{x+2} + 9 = 3^x$

Worksheet on Inequalities

Solve the following inequalities giving your answer in interval notation.

Also sketch your answer on a number line.

- $-3(r - 4) \geq 0$
- $\frac{-9+a}{15} > 1$
- $-1 > \frac{12+x}{4}$
- $6 - 4(6n + 7) \geq 122$
- $-x \geq -x + 7(x - 2)$
- $m - 2 < -8$ or $\frac{m}{8} > 1$
- $-1 < 9 + n < 17$
- $x + 8 \geq 9$ and $\frac{x}{7} \leq 1$
- $2n + 7 \geq 27$ or $3 + 3n \leq 30$
- $8r - 5 \geq 6r - 1$ or $4 + 4r \leq 3r - 3$

Worksheet on Exponents

Laws ($a, b \in \mathbb{R}$ and $n, m \in \mathbb{Z}$)

- $a^m \cdot a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $(ab)^m = a^m b^m$
- $\sqrt[n]{a^n} = a^{\frac{n}{n}}$

Exercise 1

Simplify without the use of a calculator

- $81^{\frac{1}{4}} \times 27^{-\frac{2}{3}}$
- $\frac{6^{6x} 9^{3x}}{54^{4x} \left(\frac{1}{4}\right)^{2-x}}$
- $\frac{2^{n+1}}{(2n)^{n-1}} \div \frac{4^{n+1}}{(2^{n-1})^{n+1}}$
- $\frac{18^x(2 \cdot 3^{1-x})}{2^{x-1}}$
- $125^{\frac{2}{3}} + 16^{\frac{3}{4}} - 8^{-\frac{2}{3}}$
- $\frac{9^{\frac{1}{2}x} - 3^{x-1}}{3^{x+2}}$
- $\frac{3 \cdot 2^m - 4 \cdot 2^{m+2}}{2^m - 2^{m+1}}$
- $\frac{2^{2x} - 2^x - 6}{2^{x+2}}$

Exercise 2

Solve x without the use of a calculator

- $4 \cdot 8^{x-1} = \frac{2}{16^x}$
- $3^{x+1} - 3^{x-1} = 24\sqrt{3}$
- $2^{2x+1} = -4 + 9 \cdot 2^x$
- $5^x - 24 = 5^{2-x}$
- $3^{2x} = 6 \cdot 3^{1-x} + 3$